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Wide-angle multimodal parabolic equations: modelling of directional sound propagation in stratified range-dependent fluid waveguides

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This article presents a multimodal formulation for parabolic equations (multimodal PE) that can be applied to the modelling of directional sound propagation in stratified, range-dependent fluid environments. Multimodal PE projects sound pressure onto a space spanned by a set of basis modes, which are defined in a bounded domain with the water surface and a false bottom truncating the sediment layer. This projection results in a one-way equation for the projection coefficients, with the depth operator implemented in the projection space. The proposed multimodal PE offers three advantages: (1) it provides stable solutions that analytically account for discontinuities in sound speed and density at stratified interfaces, making it well-suited to multi-layer sediment environments; (2) its computational complexity remains essentially unchanged even as the number of sediment layers increases; and (3) it exhibits strong convergence behaviour proportional to $1/M^2$, where M is the number of basis modes, enabling efficient solutions when high accuracy is required.

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Moreover, multimodal PE supports very-wide-angle directional sound propagation within a single PE simulation, without added numerical complexity compared with omnidirectional propagation. Numerical simulations validate the accuracy, efficiency and source-directivity modelling capability of the proposed approach.

1. Introduction

The modelling of underwater acoustic propagation has significant applications in marine resource exploration, sea rescue operations, anthropogenic noise mitigation, and sonar performance evaluation. Over the past two decades, extensive research has focused on the development of underwater acoustic propagation models. Prominent examples widely used in the underwater acoustics community include normal mode methods [1–4], ray tracing techniques [5–7], parabolic equations (PEs) [8–10], and fully numerical approaches such as finite element (FE) models [11–13] and spectral element models [14–16]. Among these, PEs provide a powerful tool for addressing range-dependent problems at low frequencies, where ray tracing lacks sufficient accuracy. Compared with the coupled normal mode method, PEs are computationally efficient because a one-way equation is solved rather than the full Helmholtz equation, offering a balanced trade-off between accuracy and efficiency. This makes PEs particularly well-suited for modelling long-range propagation in environments with spatially varying horizontal sound speed profiles (SSPs). For example, a recent study employed PEs to evaluate differences between an acoustically inferred sound speed map and a cross-section of sound speed interpolated from multiple conductivity, temperature, and depth profiles [17]. Furthermore, PEs have been widely used to investigate the impacts of seasonal variations in the Pacific Summer Water layer and ice cover on underwater sound propagation in the Beaufort Sea [18–21].

The PE method is derived from the paraxial approximation to the Helmholtz equation. This assumes that outgoing energy is dominant in environments with gradual horizontal variations, resulting in the so-called one-way equation [22]. The one-way equation is range-evolutionary and can thus be solved efficiently. There have been significant advances in PE-based methods in terms of capability [23,24], reliability [25,26], and accuracy [27], but, to the best of the authors' knowledge, existing approaches are limited to omnidirectional point sources. In scenarios where most underwater noise sources are directional, such as underwater vehicles and surface vessels, predictions based on PEs may exhibit significant deviations from observations and experimental data.

In principle, the PE method can simulate directional sound propagation if an appropriate initial field is provided. Obtaining such an initial field typically involves numerical modelling tools such as FE, whereby the near-field pressures of a directional source are computed along a vertical line extending from the sea surface to the outer boundary of the PE computational domain [28]. However, this approach becomes computationally expensive with increasing water depth. Moreover, phase errors accumulate because the energy of directional beams with large grazing angles couples into the propagating modes, particularly in cases involving a hard seabed or deep water and when a narrow-angle PE is used.

An alternative approach for incorporating directional sources into PEs is to integrate the sound field produced by an array of point sources [29], where the Green's function of each point source is calculated using PEs. This method, however, requires repetitive PE computations, resulting in increased computational cost. A more efficient technique was proposed by Vecherin *et al.* [30,31], who introduced a spherical wave decomposition to represent the source directivity. In this method, each spherical harmonic order is reconstructed using a set of point sources. The initial field for these point sources is readily obtained using existing PE solvers. Directional sound propagation in range-dependent ocean environments is then computed by superposing the fields generated by different spherical harmonic orders, with weighting coefficients determined by the

spherical wave decomposition. This approach is more efficient than direct numerical integration of point source fields, but still demands additional computational effort.

Previously, the authors demonstrated that directional propagation can be achieved by deriving the modal excitation of normal modes [32,33], which was validated in range-independent environments. More recently, this approach was extended to three-dimensional problems by incorporating adiabatic normal modes, enabling the solution of two-way wave equations that govern directional propagation in single-layer, unbounded spaces [34]. However, to the best of the authors' knowledge, the development of one-way wave equations for directional propagation in stratified media, where several interfaces must be addressed, remains an open problem. This direction is particularly appealing because of its potential advantages in computational efficiency and model adaptability.

The primary motivation of this study is to develop a PE model that can effectively handle density and sound speed discontinuities at stratified interfaces and enable the simulation of sources with arbitrary prescribed directivity and very wide-angle propagation. Here, we characterize the wide-angle validity of the PE by the maximum propagation angle relative to the paraxial direction for which the PE approximation maintains a prescribed accuracy (e.g. phase error below a tolerance). In this paper, we present a multimodal formulation of the PE (multimodal PE). This approach involves projecting pressures onto a space spanned by a set of basis modes, which satisfy the boundary conditions at the sea surface and a false bottom representing the outer boundary of the truncated domain. To absorb outgoing waves propagating to infinity, a perfectly matched layer (PML) [35] is incorporated. This projection yields a one-way equation for the projection coefficients, rather than for the pressures, with the depth operator implemented in the projection space. The original contributions of this work are:

- By defining a projection of the sound pressure field onto basis modes (simple sine functions satisfying the boundary conditions at the sea surface and the false bottom), we derive a PE to advance the corresponding projection coefficients along range. This formulation accommodates stratified media with multiple discontinuities in density and sound speed across interfaces within the projection space. By analytically enforcing continuity conditions at stratified interfaces, the proposed multimodal PE achieves enhanced convergence at a rate of $O(1/M^2)$, where M denotes the number of basis modes. As a result, the method remains computationally efficient even under high-accuracy requirements, and the complexity increases only marginally with the number of interfaces for a fixed projection space.
- Based on this projection, an analytical starter for sources with arbitrary prescribed directivity is derived directly from the modal excitation developed by He *et al.* [32,33]. This starter supports very-wide-angle propagation, up to nearly $\pm 90^\circ$, and integrates seamlessly with the multimodal PE framework.

2. Theory

(a) Standard PEs

Consider the standard two-dimensional Helmholtz equation in cylindrical coordinates [22]:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r} \frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + k_0^2 \varsigma(r, z)^2 p = 0, \quad (2.1)$$

where $p(r, z)$ represents the acoustic pressure, k_0 denotes the reference wavenumber associated with the reference sound speed c_0 , and ς is the index of refraction.

Here, we neglect continuous, depth-dependent density variations in the wave equation for the following reasons: (1) continuous, depth-dependent density variations are not the primary

contributors to reflected and refracted waves, as they mainly influence wave amplitudes; (2) this simplification allows analytical expressions to be obtained when constructing the depth operator in the proposed multimodal PE model, thereby improving computational efficiency. Nevertheless, the proposed theory accounts for density discontinuities at stratified interfaces and range-dependent density variations. Moreover, depth-dependent density can be incorporated through numerical integration during the construction of the depth operator, which does not pose any fundamental limitation to the proposed method.

Expressing p as a slowly varying envelope multiplied by the outgoing cylindrical wave yields [22]

$$p(r, z) = \psi(r, z) H_0^{(1)}(k_0 r), \quad (2.2)$$

where ψ is the envelope that varies slowly with range. It is evident that ψ satisfies the boundary conditions

$$\left. \begin{aligned} \psi(r, z)|_{z=0} &= 0, \\ \frac{1}{\rho_1} \frac{d\psi(r, z)}{dz} \Big|_{z=h^+} &= \frac{1}{\rho_2} \frac{d\psi(r, z)}{dz} \Big|_{z=h^-}, \end{aligned} \right\} \quad (2.3)$$

where ρ_j ($j = 1, 2$) represent the densities and h is the depth at which the water-seabed interface occurs.

Substituting equations (2.2) into (2.1), and assuming $k_0 r \gg 1$, leads to

$$\frac{\partial^2 \psi}{\partial r^2} + 2ik_0 \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} + k_0^2 [\zeta(r, z)^2 - 1] \psi = 0. \quad (2.4)$$

By applying the paraxial approximation and selecting only the outgoing wave component in equation (2.4), the one-way equation can be derived as [22]

$$\frac{\partial \psi}{\partial r} = ik_0 \left(\sqrt{\zeta^2 + \frac{1}{k_0^2} \frac{\partial^2}{\partial z^2}} - 1 \right) \psi. \quad (2.5)$$

The pseudodifferential operator $\sqrt{\zeta^2 + \frac{1}{k_0^2} \frac{\partial^2}{\partial z^2}}$ is traditionally handled using the Galerkin method, which involves discretizing the depth operator in the spatial domain. To summarize, standard PEs are based on four key assumptions: (1) the far-field approximation, where $k_0 r \gg 1$; (2) the neglect of backscattering; (3) the neglect of the commutator that appears in equation (2.4); and (4) the paraxial approximation, under which the second-order range derivative $\partial^2 \psi / \partial r^2$ in equation (2.4) is neglected.

(b) Multimodal formulation for the depth operator

The basic idea is to introduce a projection of the slowly varying function ψ , that is [36],

$$\left. \begin{aligned} \psi(r, z) &= \sum_{m=1}^M a_m(r) \phi_m(z) = \boldsymbol{\Phi}^T \mathbf{a}, \\ \phi_m(z) &= \sqrt{\frac{2}{H}} \sin \left[\frac{m\pi}{H} z \right], \end{aligned} \right\} \quad (2.6)$$

where ϕ_m are the basis modes, a_m are their coefficients, and $\boldsymbol{\Phi}$ denotes the multimodal matrix with the m -th row represented by ϕ_m . As shown in figure 1, the basis modes ϕ_m are defined as a set of orthogonal functions that satisfy pressure-release boundary conditions at both the sea surface and the false bottom. It is important to note that these basis modes do not satisfy the continuity conditions across stratified interfaces, which distinguishes them from normal modes. The projection of the acoustic pressure field onto ϕ_m is a spectral form of the Galerkin approximation, which subsequently facilitates the elimination of the continuity conditions for the envelope function ψ . In the above equation, H represents the depth of the false bottom.

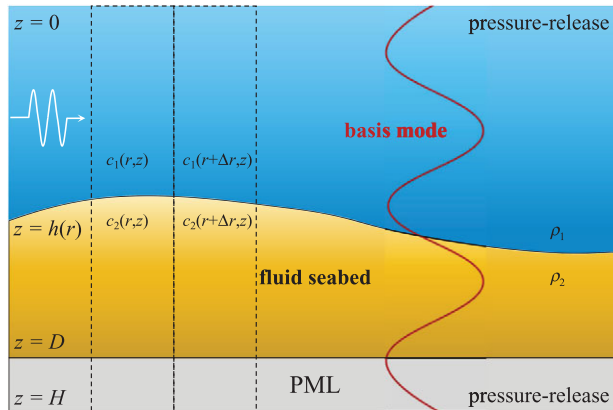


Figure 1. Schematic of the computation domain truncated by a PML, where the basis mode that satisfies pressure-release conditions at the upper ($z = 0$) and lower ($z = H$) boundaries of the bounded space is annotated by the red solid line.

Next, we convert [equation \(2.4\)](#) into a first-order evolution equation

$$\frac{\partial}{\partial r} \begin{bmatrix} \psi \\ \psi' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -[\partial_z^2 + k_0^2(\varsigma^2 - 1)] & -2ik_0 \end{bmatrix} \begin{bmatrix} \psi \\ \psi' \end{bmatrix}, \quad (2.7)$$

where ψ' represents the range derivative of ψ . Multiplying [equation \(2.7\)](#) by $\rho^{-1}\Phi$ and integrating the resulting weighted equation over the interval $0 \leq z \leq H$ gives

$$\begin{bmatrix} \mathbf{A} & 0 \\ 0 & \mathbf{A} \end{bmatrix} \begin{bmatrix} \frac{\partial}{\partial r} \mathbf{a} \\ \frac{\partial}{\partial r} \mathbf{a}' \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{A} \\ -k_0^2 \mathbf{B} & -2ik_0 \mathbf{A} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{a}' \end{bmatrix}, \quad (2.8)$$

where $\mathbf{a}'$ represents the range derivative of projection coefficient $\mathbf{a}$ and

$$\mathbf{A} = \int_0^H \frac{1}{\rho_j} \Phi \Phi^T dz, \quad j = 1, 2, \quad (2.9)$$

and

$$\mathbf{B} = \int_0^H \frac{1}{\rho_j k_0^2} \frac{\partial \Phi}{\partial z} \frac{\partial \Phi^T}{\partial z} - \frac{1}{\rho_j} (\varsigma^2 - 1) \Phi \Phi^T dz, \quad j = 1, 2. \quad (2.10)$$

[Equation \(2.8\)](#) represents a coupled-mode equation with coupling matrices defined by [equations \(2.9\)](#) and [\(2.10\)](#). It should be noted that the first term of $\mathbf{B}$ is derived via integration by parts, which reduces the second-order derivatives in [equation \(2.7\)](#) to first-order derivatives, yielding

$$\begin{aligned} \int_0^H \frac{1}{\rho_j k_0^2} \Phi \frac{\partial^2 \psi}{\partial z^2} dz &= \Phi \left(\frac{1}{\rho_1} \frac{\partial \psi}{\partial z} - \frac{1}{\rho_2} \frac{\partial \psi}{\partial z} \right) \Big|_{z=h} - \int_0^H \frac{1}{\rho_j k_0^2} \frac{\partial \Phi}{\partial z} \frac{\partial \psi}{\partial z} dz \\ &= - \int_0^H \frac{1}{\rho_j k_0^2} \frac{\partial \Phi}{\partial z} \frac{\partial \Phi^T}{\partial z} dz \mathbf{a}. \end{aligned} \quad (2.11)$$

In the above derivation, the elimination of $\Phi(h)(\rho_1^{-1} \partial_z - \rho_2^{-1} \partial_z) \psi(h)$ is based on the fact that the envelope function ψ satisfies the continuity across the water–seabed interface defined in [equation \(2.3\)](#). Evidently, the proposed multimodal PE requires no additional numerical effort to address the density discontinuity across the water–seabed interface. This is because the construction of the operator $\mathbf{B}$ explicitly incorporates the boundary condition at the interface by analytically eliminating the term $\Phi(h)(\rho_1^{-1} \partial_z - \rho_2^{-1} \partial_z) \psi(h)$. We further note that, although the chosen sine basis functions do not automatically satisfy the continuity of normal velocity at the

water–seabed interface, their orthogonality and completeness on the bounded interval make them suitable for expanding the true eigenfunctions of the acoustic waveguide, which do satisfy this continuity condition (see Appendix for this derivation). In particular, because the waveguide is bounded by two pressure-release boundaries, any normal mode that enforces continuity of normal velocity across a fluid–fluid interface within the waveguide can be represented as a linear superposition of these sine functions. This observation underpins the rationale for the above elimination step.

While the number of sine basis functions required for convergence may exceed the number of propagating normal modes (see the supplementary material for details), we emphasize that the sine basis is range-independent in a fixed computation domain, and the resulting depth operator is partly analytical, both of which substantially reduce computational complexity and enable the present multimodal PE formulation.

Equation (2.8) can be further simplified as

$$\frac{\partial}{\partial r} \begin{bmatrix} \mathbf{a} \\ \mathbf{a}' \end{bmatrix} = \begin{bmatrix} 0 & \mathbf{I} \\ -k_0^2 \mathbf{A}^{-1} \mathbf{B} & -2ik_0 \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{a}' \end{bmatrix}. \quad (2.12)$$

This is essentially a first-order evolution equation along r for the coefficients of the basis modes. After eliminating $\mathbf{a}'$, equation (2.12) can be rewritten as

$$\left[\frac{\partial^2}{\partial r^2} + 2ik_0 \frac{\partial}{\partial r} + k_0^2 \left(\sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} - \mathbf{I} \right) \right] \mathbf{a} = 0. \quad (2.13)$$

Factorizing equation (2.13) into outgoing and incoming wave components, and neglecting the commutator term, results in

$$\left(\frac{\partial}{\partial r} + ik_0 \mathbf{I} - ik_0 \sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} \right) \left(\frac{\partial}{\partial r} + ik_0 \mathbf{I} + ik_0 \sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} \right) \mathbf{a} = 0. \quad (2.14)$$

Retaining the outgoing wave component in equation (2.14) yields the one-way equation formulated by the multimodal approach:

$$\frac{\partial \mathbf{a}}{\partial r} = ik_0 (\sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} - \mathbf{I}) \mathbf{a}. \quad (2.15)$$

Equation (2.15) is essentially the multimodal formulation of equation (2.5), with the marching variable replaced by the projection coefficients $\mathbf{a}$ rather than the unknown pressure envelope. This formulation facilitates the derivation of an analytical initial field for arbitrary directional sources, as discussed later in this paper. Additionally, this equation can be regarded as the one-way form of coupled-mode equations for basis modes.

The solution to equation (2.15) can be formally written as

$$\mathbf{a}(r + \Delta r) = e^{ik_0 \Delta r (\sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} - \mathbf{I})} \mathbf{a}(r). \quad (2.16)$$

Equation (2.16) can be solved directly, but it imposes a strict constraint on the range step Δr . To maintain stability, the marching scheme often requires $\Delta r \approx \lambda/10$. This restriction can be relaxed, approximately to $\lambda/2$, by reformulating the marching scheme, for example, Petrov *et al.* [37] derived a matrix-WKB method for coupled-mode propagation in underwater acoustics. In the present work, we instead approximate the square root of the depth operator in equation (2.16) by a rational expansion, i.e. the split-step Padé solution [38], which permits substantially larger Δr while maintaining numerical stability. This method enables the square-root operator in equation (2.16) to be expanded into a sequential product of rational operators:

$$e^{ik_0 \Delta r (\sqrt{\mathbf{A}^{-1} \mathbf{B} + \mathbf{I}} - \mathbf{I})} = \prod_{l=1}^L \frac{1 + \alpha_{l,L} \mathbf{A}^{-1} \mathbf{B}}{1 + \beta_{l,L} \mathbf{A}^{-1} \mathbf{B}}, \quad (2.17)$$

where $\alpha_{l,L}$ and $\beta_{l,L}$ represent the coefficients of a Padé approximant of order L . The method for calculating these coefficients has been described by Antoine *et al.* [39,40], and is not repeated

here. It is worth noting that range-dependent density variations can be incorporated into the multimodal PE because the sound field is advanced using a marching scheme that partitions the environment into range slices, each of which may have distinct density and sound speed.

(c) Construction of the operators $\mathbf{A}$ and $\mathbf{B}$

The depth operator $\mathbf{A}^{-1}\mathbf{B}$ can be constructed using the projection given by [equation \(2.5\)](#). We use a PML to truncate the half-space, which extends from $z = D$ to $z = H$ [35], i.e.,

$$\hat{z} = z + i \int_0^z \sigma(\varepsilon) d\varepsilon. \quad (2.18)$$

Here, $\sigma(\varepsilon)$ is the damping function, with the dimensionless coordinate $\varepsilon = (z - D)/(H - D)$. In this study, we use a second-degree polynomial damping function [41,42]. From [equation \(2.18\)](#), we obtain

$$\left. \begin{aligned} \frac{\partial}{\partial \hat{z}} &= \frac{1}{s(z)} \frac{\partial}{\partial z}, \\ d\hat{z} &= s(z)dz, \end{aligned} \right\} \quad (2.19)$$

and

where

$$\left. \begin{aligned} s(z) &= 1, \quad 0 < z < D, \\ s(z) &= 1 + i\sigma(z), \quad D < z < H. \end{aligned} \right\} \quad (2.20)$$

Substituting [equations \(2.18\)](#) into [\(2.9\)](#) and dividing the integral into three parts according to the locations of the water–seabed interface and the inner boundary of the PML yields:

$$\begin{aligned} \mathbf{A} &= \frac{1}{\rho_1} \int_0^h \boldsymbol{\Phi} \boldsymbol{\Phi}^T dz + \frac{1}{\rho_2} \int_h^D \boldsymbol{\Phi} \boldsymbol{\Phi}^T dz + \frac{1}{\rho_2} \int_D^H s(z) \boldsymbol{\Phi} \boldsymbol{\Phi}^T dz \\ &= \mathbf{A}_1 + \mathbf{A}_2 + \frac{1}{\rho_2} \int_D^H s(z) \boldsymbol{\Phi} \boldsymbol{\Phi}^T dz, \end{aligned} \quad (2.21)$$

where the first two terms can be expressed analytically as

$$\left. \begin{aligned} \mathbf{A}_1 &= -\frac{h}{\rho_1 H} \left[\text{sinc} \frac{h(m+n)}{H} - \text{sinc} \frac{h(m-n)}{H} \right] \\ \text{and } \mathbf{A}_2 &= -\frac{1}{\rho_2 H} \left[D \text{sinc} \frac{D(m+n)}{H} - h \text{sinc} \frac{h(m+n)}{H} - D \text{sinc} \frac{D(m-n)}{H} + h \text{sinc} \frac{h(m-n)}{H} \right]. \end{aligned} \right\} \quad (2.22)$$

Similarly, the operator $\mathbf{B}$ can be written as

$$\mathbf{B} = \mathbf{C} - \mathbf{E}, \quad (2.23)$$

where

$$\begin{aligned} \mathbf{C} &= \frac{1}{\rho_1 k_0^2} \int_0^h \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} dz + \frac{1}{\rho_2 k_0^2} \int_h^D \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} dz + \frac{1}{\rho_2 k_0^2} \int_D^H \frac{1}{s(z)} \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} dz \\ &= \mathbf{C}_1 + \mathbf{C}_2 + \frac{1}{\rho_2 k_0^2} \int_D^H \frac{1}{s(z)} \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} dz, \end{aligned} \quad (2.24)$$

where the two analytical terms, $\mathbf{C}_1$ and $\mathbf{C}_2$, are given by

$$\left. \begin{aligned} \mathbf{C}_1 &= -\frac{\pi^2 h(mn)}{\rho_1 H^3} \left[\text{sinc} \frac{h(m+n)}{H} + \text{sinc} \frac{h(m-n)}{H} \right] \\ \text{and } \mathbf{C}_2 &= -\frac{\pi^2 (mn)}{\rho_2 H^3} \left[D \text{sinc} \frac{D(m+n)}{H} - h \text{sinc} \frac{h(m+n)}{H} + D \text{sinc} \frac{D(m-n)}{H} - h \text{sinc} \frac{h(m-n)}{H} \right]. \end{aligned} \right\} \quad (2.25)$$

Because the SSP is represented as a depth-dependent variable at each step size, i.e. $c(r, z)$ or $\varsigma(r, z)$, the integral for the matrix $\mathbf{E}$ must be calculated numerically:

$$\mathbf{E} = - \int_0^H \frac{1}{\rho_j} (\varsigma^2 - 1) \boldsymbol{\Phi} \boldsymbol{\Phi}^T d\hat{z}, \quad j = 1, 2. \quad (2.26)$$

(d) Analytical starter for the source with arbitrary directivities

The variable that advances in range is the projection coefficient rather than the acoustic pressure. Hence, the selection of the initial field for solving the PE differs from that of existing methods.

For a directional source, the analytical expression for modal excitation was derived by He *et al.* [32] based on normal-mode theory. The derivation begins with the Huygens integral of a set of point sources that replace the radiation from a volume source, with the Green's function expressed in terms of normal modes. Next, by interchanging the order of the modal summation and the integral, an analytical expression is derived for the modal excitation of a directional source. This simplification is achieved by introducing the Fraunhofer approximation and a Taylor series expansion of the integral kernel. The solution is ultimately expressed as a sum over normal modes, with the modal amplitude determined by the source directivity, which in turn depends on the propagation angle of the modal plane waves. Here, we present only the final expression without providing the detailed derivation [32,33]. The analytical expression for the modal excitation of a directional source is

$$\psi(r_0, z) = \boldsymbol{\Upsilon}(z)^T \mathbf{b}(r_0), \quad (2.27)$$

where $\boldsymbol{\Upsilon}(z)$ denotes the normal modes, and their amplitudes are given by

$$\mathbf{b}(r_0) = 2\pi \boldsymbol{\Lambda}(r_0) \mathbf{d}(z_s). \quad (2.28)$$

In the above equation, the diagonal matrix $\boldsymbol{\Lambda}(r_0)$ is given by

$$\boldsymbol{\Lambda}(r_0) = \begin{bmatrix} \frac{e^{ik_{r,1}r_0}}{k_{r,1}} & 0 & \cdots & 0 \\ 0 & \frac{e^{ik_{r,2}r_0}}{k_{r,2}} & \cdots & 0 \\ \vdots & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{e^{ik_{r,M}r_0}}{k_{r,M}} \end{bmatrix}, \quad (2.29)$$

and the vector $\mathbf{d}(z_s)$ is

$$\mathbf{d}(z_s) = \frac{1}{2} \begin{bmatrix} -i \frac{d\Upsilon(z)}{dz} \Big|_{z=z_s} \frac{1}{k_{z,1}} \mathcal{D}^- + \Upsilon(z_s) \mathcal{D}^+ \\ -i \frac{d\Upsilon(z)}{dz} \Big|_{z=z_s} \frac{1}{k_{z,2}} \mathcal{D}^- + \Upsilon(z_s) \mathcal{D}^+ \\ \vdots \\ -i \frac{d\Upsilon(z)}{dz} \Big|_{z=z_s} \frac{1}{k_{z,M}} \mathcal{D}^- + \Upsilon(z_s) \mathcal{D}^+ \end{bmatrix}, \quad (2.30)$$

where k_r denotes the eigenvalue of the normal mode, k_z denotes the vertical wavenumber. The terms $\mathcal{D}^-$ and $\mathcal{D}^+$, which depend on the source directivity, are defined as

$$\left. \begin{aligned} \mathcal{D}_m^- &= \mathcal{D}(\varphi_m) - \mathcal{D}(-\varphi_m), \\ \mathcal{D}_m^+ &= \mathcal{D}(\varphi_m) + \mathcal{D}(-\varphi_m), \end{aligned} \right\} \quad (2.31)$$

and

where $\mathcal{D}$ represents the source directivity and $\varphi_m = \arctan(k_{z,m}/k_{r,m})$ is the propagation angle of the m -th normal mode, with $k_{z,m} = \sqrt{\omega^2/c^2(z_s) - k_{r,m}^2}$.

Following the modal projection method proposed by He *et al.* [32], the normal modes Υ can be projected onto the space spanned by the basis modes Φ using the projection

$$\Upsilon^T = \Phi^T \mathbf{W}, \quad (2.32)$$

where $\mathbf{W}$ is the projection coefficient matrix, obtained through the eigenvalue decomposition of $\sqrt{\mathbf{A}^{-1}\mathbf{B}}$ [32]. Additionally, the envelope ψ can be written as

$$\psi = \Phi^T \mathbf{a} = \Upsilon^T \mathbf{b}. \quad (2.33)$$

Substituting equation (2.33) into equation (2.32) gives

$$\mathbf{W}\mathbf{b} = \mathbf{a}. \quad (2.34)$$

Equation (2.34) essentially describes the projection of the normal-mode amplitudes $\mathbf{b}$ onto the coefficients of the basis modes $\mathbf{a}$. Using this projection and equation (2.28), one can obtain the initial field for the multimodal PE due to a directional source

$$\mathbf{a}(r_0) = \mathbf{W}\mathbf{b}(r_0) = 2\pi\mathbf{W}\mathbf{A}(r_0)\mathbf{d}(z_s), \quad (2.35)$$

Using equation (2.35) for the initial field yields directional sound propagation. Details regarding the solution of the eigenproblem associated with k_r , together with the modal function Υ and projection matrix $\mathbf{W}$, can be found in appendix A.

(e) Energy conservation

When addressing a sloping seafloor, the PE implementation encounters a fundamental issue with energy conservation, resulting in energy loss for upslope propagation and energy gain for downslope propagation. As suggested in a previous study [43], matching $\psi/\sqrt{\rho c}$ (the density-reduced pressure) provides an effective solution to the energy-conservation problem. This approach is therefore adopted here. The proposed multimodal PE operates in the projection space, where the variable advanced in range is the coefficient of the basis mode, rather than the pressure in the spatial domain. However, the pressure can be derived using the transformation given in equation (2.6). The projection coefficient for the density-reduced pressure $\psi/\sqrt{\rho c}$ can be obtained by applying the inverse transformation of equation (2.5) based on orthogonality:

$$\mathbf{a} = \int \Phi^T \psi \, dz. \quad (2.36)$$

To summarize, the initial projection coefficient $\mathbf{a}(r)$ is first used to reconstruct the pressure envelope $\psi(r)$ through equation (2.6). The pressure envelope is then divided by $\sqrt{\rho(r)c(r)}$. The inverse transformation is subsequently applied to obtain the projection coefficient $\mathbf{a}'(r)$ for the density-reduced pressure using equation (2.36). The projection coefficient $\mathbf{a}'(r)$ is then advanced in the projection space using the propagator defined by equation (2.17) to obtain $\mathbf{a}'(r + \Delta r)$. Finally, $\mathbf{a}'(r + \Delta r)$ is used to recover the pressure envelope by applying equation (2.6) and then multiplying by $\sqrt{\rho(r + \Delta r)c(r + \Delta r)}$.

(f) Extending the multimodal PE to multi-layer seabed scenarios

Since no additional numerical effort is required to address the density discontinuity across the water–seabed interface, the multimodal PE can be extended to more general scenarios, in which the seabed is multi-layer. Consider a waveguide consisting of J layers, each of which has a density of ρ_j and a sound speed of c_j ($j = 1, 2, \dots, J$). The interface above the j -th layer is located at $z = h_j$, where $h_1 = 0$ and $h_{J+1} = D$. In the case of a multi-layer waveguide, the propagator remains the

same as that given in [equation \(2.17\)](#), with the operators **A** and **B** modified straightforwardly as

$$\mathbf{A} = \sum_{j=1}^J \frac{1}{\rho_j} \int_{h_j}^{h_{j+1}} \boldsymbol{\Phi} \boldsymbol{\Phi}^T \, dz + \frac{1}{\rho_J} \int_D^H s(z) \boldsymbol{\Phi} \boldsymbol{\Phi}^T \, dz \quad (2.37)$$

and

$$\begin{aligned} \mathbf{B} = & \sum_{j=1}^J \frac{1}{\rho_j k_0^2} \int_{h_j}^{h_{j+1}} \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} \, dz + \frac{1}{\rho_J k_0^2} \int_D^H \frac{1}{s(z)} \frac{\partial \boldsymbol{\Phi}}{\partial z} \frac{\partial \boldsymbol{\Phi}^T}{\partial z} \, dz \\ & - \sum_{j=1}^J \int_{h_j}^{h_{j+1}} \frac{1}{\rho_j} (\zeta_j^2 - 1) \boldsymbol{\Phi} \boldsymbol{\Phi}^T \, dz - \int_D^H \frac{s(z)}{\rho_J} (\zeta_J^2 - 1) \boldsymbol{\Phi} \boldsymbol{\Phi}^T \, dz. \end{aligned} \quad (2.38)$$

It should be noted that **A** and the first term in **B** can be expressed analytically, analogous to [equations \(2.22\)](#) and [\(2.25\)](#). Additionally, the elimination of $\boldsymbol{\Phi}(h_j)(\rho_j^{-1} \partial_z - \rho_{j+1}^{-1} \partial_z) \psi(h_j)$ ($j = 2, 3, \dots, J$), which represents the continuity of normal velocity in ψ across the interface at $z = h_j$, is based on the fact that the envelope $\psi(h_j)$ automatically satisfies the boundary condition across each interface. Evidently, for a given total depth H , increasing the number of layers would not significantly increase the computational complexity as basis modes remain the same.

3. Numerical simulations

Next, we present a series of numerical simulations to demonstrate the capability of multimodal PE in modelling both omnidirectional and directional sound propagation in range-dependent underwater environments. Subsequently, we consider a case involving a multi-layer seabed to illustrate the method's ability to accurately handle density discontinuities at stratified interfaces. In these simulations, the default value of the number of basis modes M is set to $2Hf/c_0$, and the Padé approximation order is fixed at 8. Numerical tests suggest that $M = M_c = 2Hf/c_0$ provides accurate results with an acceptable runtime. This value is consistent with the number of propagating modes in the projection domain with two pressure-release boundaries. These parameters remain constant across all simulations unless otherwise specified. For comparison, the range-dependent acoustic model (RAM) [44] employs the same Padé approximation order and Δr as multimodal PE. [Table 1](#) summarizes the simulation parameters for each case, including the frequency, water (c_1 and ρ_1) and seabed (c_b and ρ_b) properties, and the range step size.

(a) Model validation

(i) ASA benchmark wedge

The multimodal PE model is validated through comparisons with the reference solution provided by RAM [44]. The test scenario is the ASA benchmark wedge, which assumes a fluid seabed with an incline of 2.86° . We consider an iso-velocity water column with a point source positioned at a depth of 100 m, where the water depth is 200 m. [Figure 2](#) illustrates the simulation setup. The remaining simulation parameters are summarized in [table 1](#).

To achieve comparable accuracy with multimodal PE (an error level of 10^{-2}), the depth discretization in RAM is set to 867 and 8751 for the 25-Hz and 250-Hz cases, respectively. The sound fields predicted by multimodal PE at 25 Hz and 250 Hz are displayed in [figure 3](#). A detailed comparison with the RAM results is provided in [figure 4](#), where the transmission loss (TL) is plotted as a function of range at a receiver depth of 30 m. In both cases, there is strong agreement between multimodal PE and RAM. RAM exhibits a discontinuity at the point where sound waves begin to penetrate the seabed in the 25-Hz case. In contrast, the multimodal PE model provides a smooth transition across the water–seabed interface because it rigorously satisfies the boundary conditions by eliminating the continuity conditions in [equation \(2.11\)](#) during integration by parts.

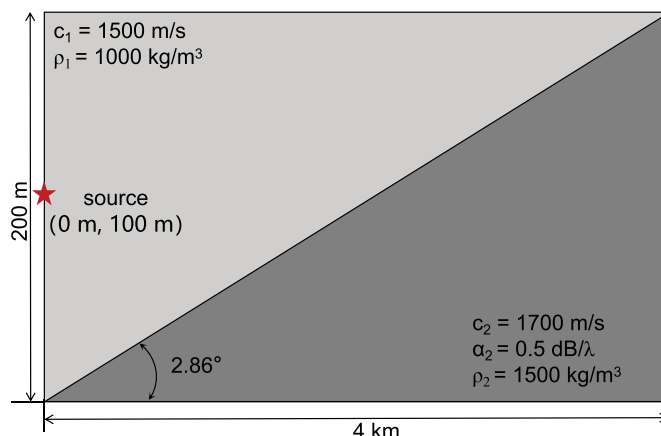


Figure 2. Schematic of the simulation setup for the ASA benchmark wedge, where a fluid seabed with an incline of 2.86° is considered. The red pentagram represents the position of the source.

Table 1. Simulation parameters: frequency, water, seabed, and range step size for each case.

	ASA wedge	range-dependent	Downslope	Seamount	Multilayer seabed
f , Hz	25/250	50	1000	230	100
ρ_1 , kg m^{-3}	1000	1000	1000	1000	1000
c_1 , m/s	1500	figure 5	1500	figure 12c	1500
ρ_b , kg m^{-3}	1500	1500	2000	1500	layer1: 1500 layer2: 2200 layer3: 2700
c_b , m/s	1700	1700	2200	1800	layer1: 1700 layer2: $2000 + 0.25 r$ layer3: 3000
α_b , dB/ λ	0.5	0.5	0.2	0.5	layer1: 0.5 layer2: 0.1 layer3: 0.1
Δr , m	3	15	2	5	3

(ii) Range-dependent benchmark

The next example examines a range-dependent scenario, established as a benchmark in the RAM handbook [44]. A schematic of the simulation setup is presented in figure 5, depicting a downslope seabed. The SSP is divided into two distinct regions: a surface duct in the upper water column for $r < 25$ km and a homogeneous water column for $r > 25$ km. The downslope starts where the water depth is 200 m, extending to where the seabed becomes flat at 400 m. The point source is placed at $z = 50$ m. The remaining simulation parameters are summarized in table 1.

Figure 6 compares the sound fields computed by RAM and multimodal PE. For RAM, the depth discretization is set to 2001 points to ensure accuracy comparable to that of multimodal PE. The results show excellent agreement, capturing both the downslope propagation and the transition behaviour at $r = 25$ km caused by the spatially varying SSP. These results underscore

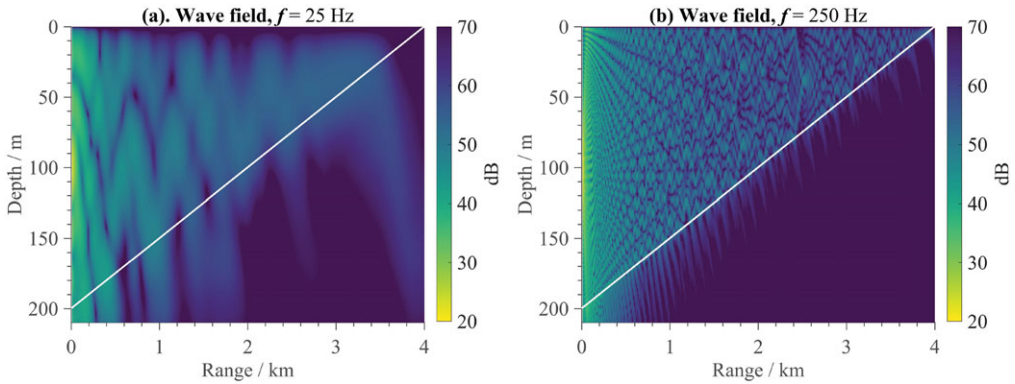


Figure 3. Sound field produced by the point source in the wedge case at (a) 25 Hz and (b) 250 Hz.

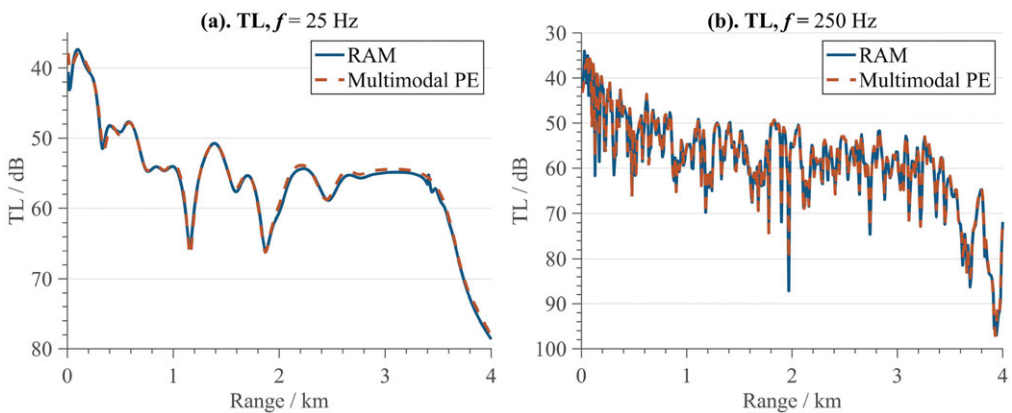


Figure 4. TL comparisons with the reference solution provided by RAM for a receiver depth of 30 m at (a) 25 Hz and (b) 250 Hz.

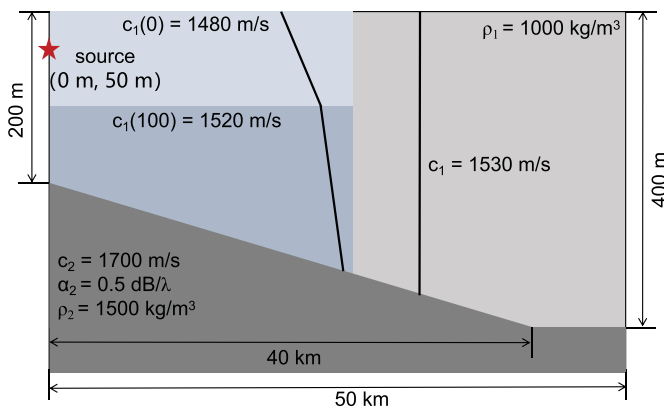


Figure 5. Schematic of the simulation setup in the range-dependent benchmark case, where a downslope seabed and a range-dependent sound speed profile are considered. The red pentagram represents the position of the source.

the effectiveness of multimodal PE in handling complex underwater environments with range dependence.

In the multimodal PE approach, the order of the propagator matrix can be reduced by retaining only the basis modes that significantly contribute to the sound field. This is accomplished by discarding the last several rows and columns of the depth operator, and keeping only the dominant propagating modes. Figures 6c,d present the results calculated using the first 20 basis

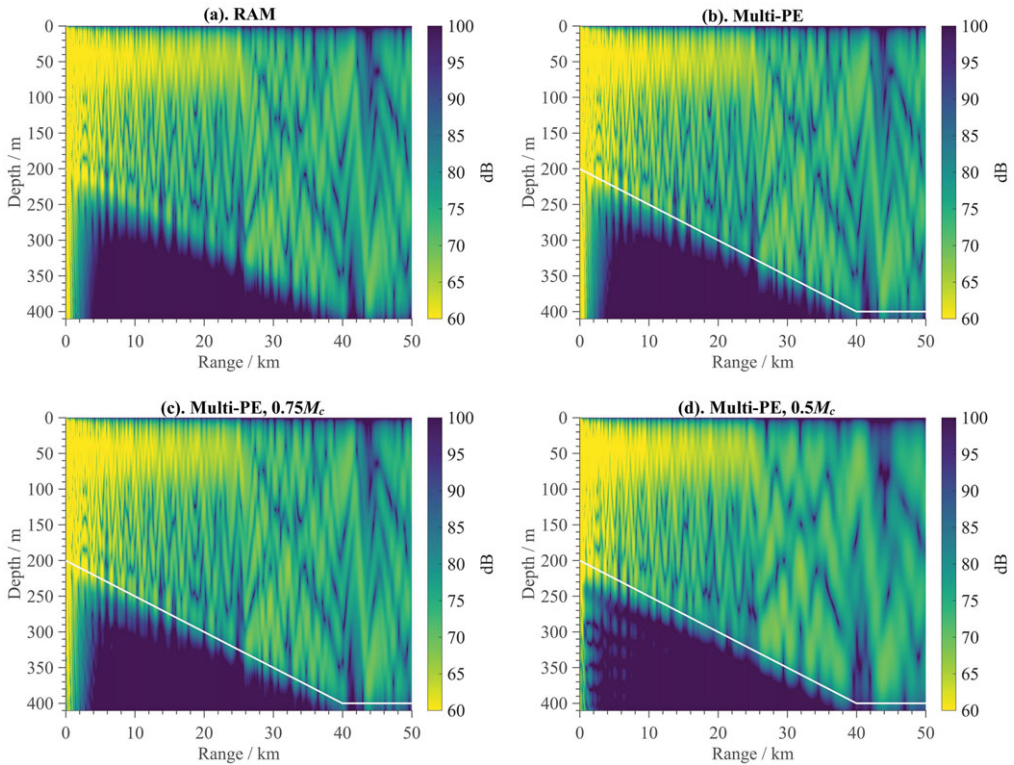


Figure 6. Sound propagation in the range-dependent benchmark case, where a downslope seabed and a spatially varying sound speed profile are considered. (a) RAM and (b) proposed multimodal PE. For comparison purposes, the results calculated using multimodal PE with the reduced-size depth operator are given in (c) and (d).

modes ($0.75M_c$) and 14 basis modes ($0.5M_c$), respectively. The results with the reduced-order operator exhibit TL contours that are very similar to the reference solution. However, as the order of the operator decreases, the interference caused by high-order modes gradually disappears. A more detailed TL comparison at a receiver depth of 50 m (figure 7) demonstrates that the overall trend of the energy distribution is accurately captured with a reduced-order operator. When reduced to 20 basis modes, the sound field remains relatively accurate, showing only a slight shift in the interference pattern compared to the reference solution. Reducing the operator order undoubtedly improves numerical efficiency, but a balance between accuracy and efficiency is crucial. Based on these results, the default value $M_c = 2fH/c_0$ (27 basis modes) is recommended as a suitable compromise, ensuring acceptable accuracy with improved computational efficiency.

(b) Convergence test and numerical efficiency

After benchmarking the proposed multimodal PE model, we analyze its convergence by evaluating the relative errors in the range-dependent benchmark case. For fairness, the RAM model was fully implemented in MATLAB [45] to take advantage of CPU multithreading instead of using the *.mex* binaries. Both models were implemented on the same computer, equipped with an Intel 13900K CPU and 128 GB of RAM. Clearly, the depth operator for RAM is implemented in the spatial domain, whereas that for multimodal PE is discretized in the projection domain. We define the relative errors as

$$\text{error} = \frac{1}{Q} \sum_q \frac{|p_q^{i+1} - p_q^i|}{|p_q^{i+1}|} \times 100\%, \quad (3.1)$$

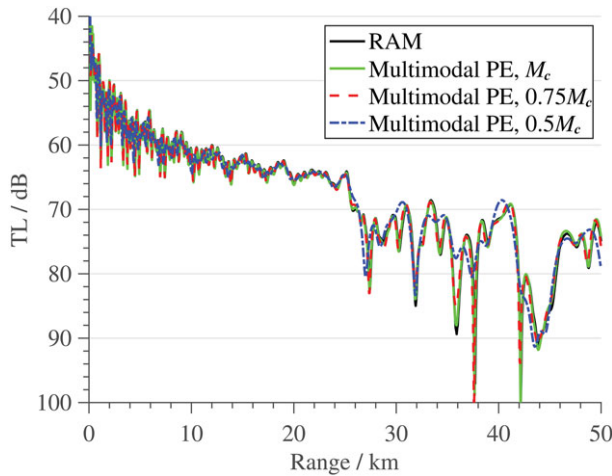


Figure 7. TL in the range-dependent benchmark case with a receiver depth of 50 m. The solid black and green lines represent the results computed by RAM and multimodal PE, respectively. The dashed red line and the dot-dashed blue line are the results with the reduced-order depth operator.

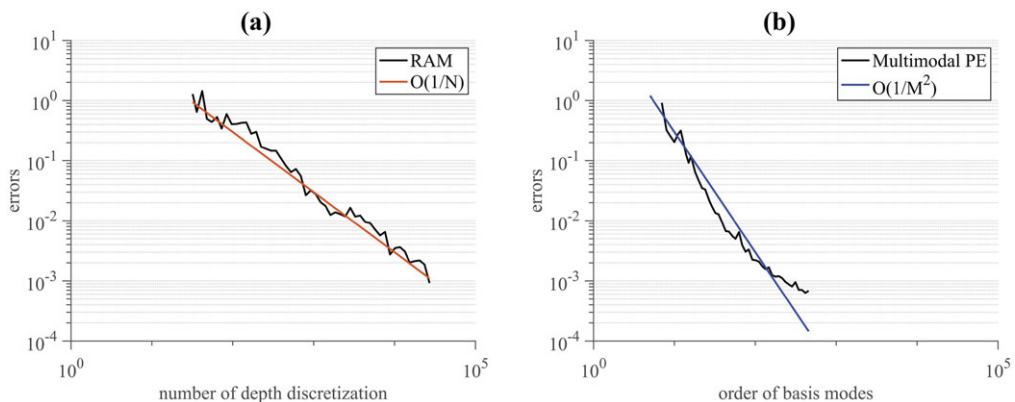


Figure 8. Convergence curves for (a) RAM as a function of the number of depth discretization points and (b) multimodal PE as a function of the order of basis modes.

where the subscript q indicates the q -th field point and Q is the total number of field points. Here, the numerator $p_q^{i+1} - p_q^i$ represents the difference between the pressures computed using a finer step size ($i + 1$) and a coarser step size (i). For RAM, this difference was evaluated with increasing numbers of depth points N , whereas for multimodal PE it was compared using increasing basis mode numbers M . Although the same error metric is evaluated using different quantities (N and M), model accuracy (convergence) of RAM and multimodal PE are equivalent when a same desired error is reached. Therefore, the error metric defined in equation (3.1) can be used to evaluate the equivalent model accuracy between these two models. Additionally, due to the nature of model reduction, M is typically smaller than N .

Figure 8 illustrates the convergence curves for RAM as a function of the number of depth discretization points N and for multimodal PE as a function of the number of basis modes M . The pressures along $z = 50$ m have been used to calculate the relative errors, with a step size of 15 m applied to both methods. The error curve for RAM decreases at a rate proportional to $O(1/N)$. In contrast, the convergence of the proposed multimodal PE follows a trend proportional to

Table 2. Comparison of computation times using the two models.

case name	error level	RAM	multimodal PE
ASA wedge (25 Hz)	10^{-1}	0.1 s	0.2 s
ASA wedge (25 Hz)	10^{-2}	0.6 s	0.4 s
ASA wedge (25 Hz)	10^{-3}	6.4 s	1.8 s
range-dependent benchmark	10^{-1}	0.6 s	1.4 s
range-dependent benchmark	10^{-2}	5 s	4.4 s
range-dependent benchmark	10^{-3}	95.5 s	26.8 s

$O(1/M^2)$. These results demonstrate that, for the same level of accuracy, the proposed multimodal PE converges faster than RAM.

To further demonstrate the numerical efficiency of multimodal PE, we compare the computation times of the two models. In this case, N and M were selected to achieve equivalent target errors based on the convergence study. Three error levels are used to evaluate how each model performs as the accuracy requirement increases. Table 2 presents the computation times for the two models under two different scenarios. The results indicate that multimodal PE is slightly slower than RAM at an error level of 10^{-1} . However, as higher accuracy is demanded, multimodal PE becomes more efficient due to its significantly faster convergence. These findings demonstrate the strong numerical convergence of multimodal PE, indicating that it becomes computationally efficient when higher accuracy is required.

In addition to the number of basis modes, the range step size Δr critically influences computational efficiency. Because the exponential propagator is approximated by a split-step Padé scheme, our multimodal PE can stably use large steps, up to 20λ , with acceptable error (see the supplementary material for details). Nevertheless, for general underwater environments, we recommend starting with $\Delta r \in [\lambda/2, \lambda]$, then performing a brief convergence test to satisfy the desired accuracy. The optimal choice of Δr is environment-dependent and will vary from case to case [22].

(c) Sound field produced by a directional source

(i) Double pistons in the downslope propagation scenario

Next, we evaluate the ability of multimodal PE to simulate sound propagation from a directional source. The source directivity is modelled as a pair of pistons with opposite elevation angles (up-going and down-going). The directivity pattern is given by

$$\mathcal{D}(\varphi) = \frac{2J_1[kr_a \sin(\varphi - \varphi_s)]}{kr_a \sin(\varphi - \varphi_s)} + \frac{2J_1[kr_a \sin(\varphi + \varphi_s)]}{kr_a \sin(\varphi + \varphi_s)}, \tag{3.2}$$

where $r_a = 10$ m is the radius and φ_s denotes the orientation angle of the piston. Here, $\varphi = 0^\circ$ indicates that the dominant radiation beam is emitted from the piston at a grazing angle of 0° .

One might argue that directional sound propagation can be achieved by RAM through the incorporation of a modal starter with an appropriate weighting function. Therefore, we compare multimodal PE with RAM that adopts such a weighted modal starter. To account for source directivity in RAM, a modal starter must be used, with its detailed formulation provided in [22]. The weighting function for each mode is then computed using equation (2.28).

We consider a downslope propagation scenario where the water depth at the source location is 100 m and increases linearly with range at a slope of $\arctan(3/40)$. The directional source is placed at a depth of 50 m. In RAM, the depth step is $\Delta z = 0.02$ m. The remaining simulation parameters are summarized in table 1.

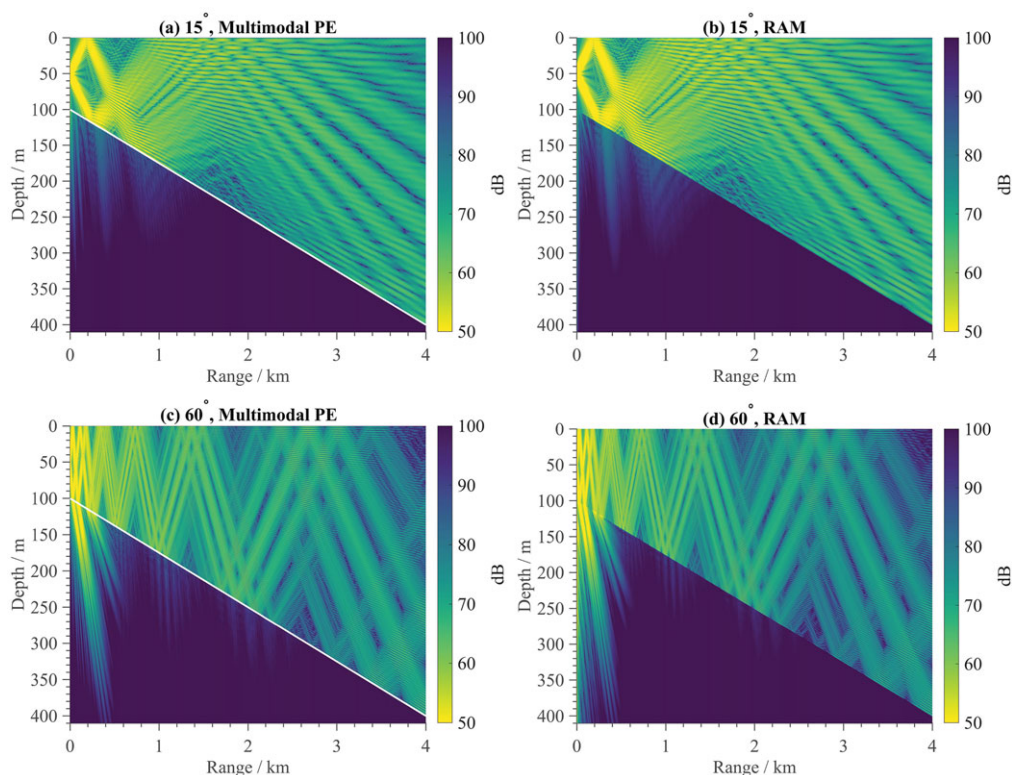


Figure 9. Sound propagation from a directional source modelled as double pistons with opposite elevation angles. (a–b) Correspond to an elevation angle of 15° , while (c) and (d) correspond to 60° . The left and right columns show results from multimodal PE and RAM, respectively.

Figure 9 compares the TLs computed by the two models. Overall, the TL contours are nearly identical, with minor discrepancies near the source region. For an elevation angle of 15° , RAM predicts less near-field leaky energy than multimodal PE, resulting in stronger far-field energy. In contrast, at a steeper angle of 60° , RAM exhibits more pronounced near-field leakage, leading to a noticeable reduction in far-field energy compared with multimodal PE.

To further examine the differences between the two models in handling directional propagation, figure 10 presents TL curves at a receiver depth of 50 m. The figure includes RAM results with coarse vertical discretization ($\Delta z = 0.2$ m) and multimodal PE results with a larger number of basis modes ($M = 3Hf/c_0$). It is evident that multimodal PE yields consistent results regardless of the number of basis modes when using $\Delta z = 0.2$ m, for both shallow and steep source grazing angles. In contrast, RAM becomes unstable at $\Delta z = 0.2$ m, exhibiting a slight phase shift at 15° and significant amplitude errors at 60° . These amplitude differences, consistent with figure 9, are attributed to varying levels of near-field energy leakage. When a point source is used instead of a directional one, both models produce nearly identical results with no significant amplitude deviation (figure 11), indicating that the design of the modal starter in RAM is crucial for accurate directional modelling, especially in downslope environments [22].

It should be noted that, when employing the modal starter in RAM, both the discrete modal summation and the high-order modes representing the continuous spectrum were considered. Therefore, a normal mode algorithm capable of accurately capturing the continuous spectrum is required for integration with RAM. Careful treatment of the continuous spectrum is essential because: (a) components of the continuous spectrum can couple into the propagating modes in downslope environments; (b) sharp source directivities with large grazing angles necessitate

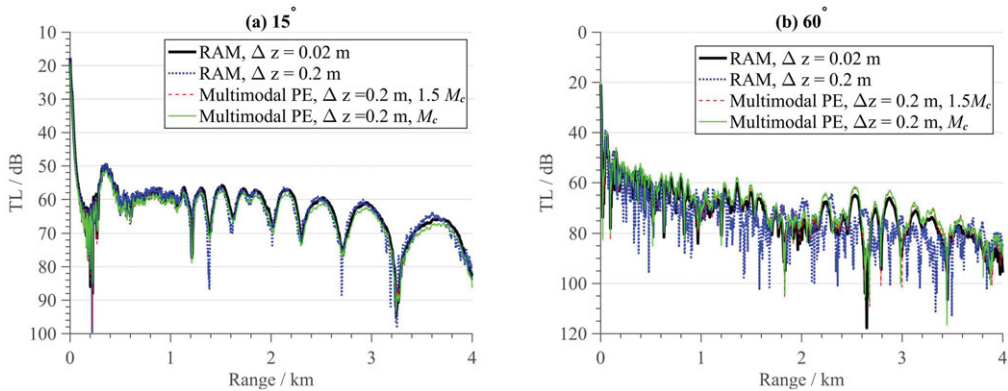


Figure 10. TLs at a receiver depth of 50 m calculated by RAM with different depth discretizations (solid black: $\Delta z = 0.02$ m; dashed blue: $\Delta z = 0.2$ m) and multimodal PE with different numbers of basis modes (dashed red: $1.5M_c$ basis modes; solid red: M_c basis modes, both with $\Delta z = 0.2$ m). The left and right columns show results for elevation angles of 15° and 60° , respectively.

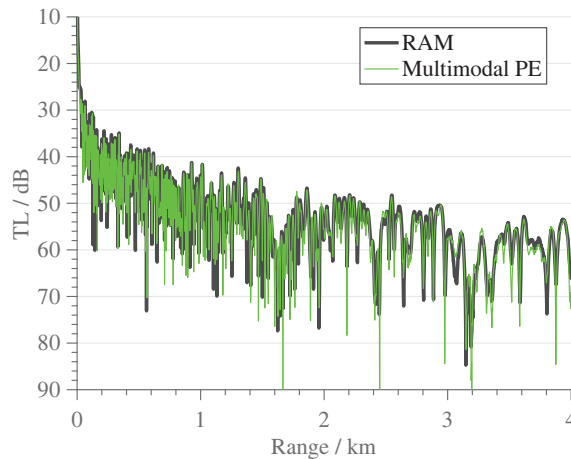


Figure 11. TLs at a receiver depth of 50 m for an omnidirectional source, computed by modal-starter RAM and multimodal PE.

contributions from the continuous spectrum, which may couple into the discrete spectrum in the far field, particularly when the seabed is sufficiently hard.

(ii) Source with an analytical directivity in the Dickins seamount scenario

We further consider directional propagation across the Dickins Seamount [22]. The seamount initially blocks propagation along deep refracted paths within the first 20 km. It has an approximate slope of 14° , and its summit lies at a depth of 500 m below the sea surface. The remaining simulation parameters are summarized in table 1. The directional source is placed at a depth of 1 km, and its directivity is given by

$$\mathcal{D}(\varphi) = -i \cos\left(2\varphi + \frac{\pi}{6}\right) \left[\sin\left(2\varphi + \frac{\pi}{6}\right) + 0.25 \right] + \sin\left(\varphi + \frac{\pi}{6}\right) + 0.55 \cos\left(6\varphi + \frac{\pi}{6}\right) \sin\left(6\varphi + \frac{\pi}{6}\right). \quad (3.3)$$

Figure 12b shows the sound field generated by the directional source over both short- and long-range propagation distances. At short ranges, a pronounced beam pattern that coincides with the

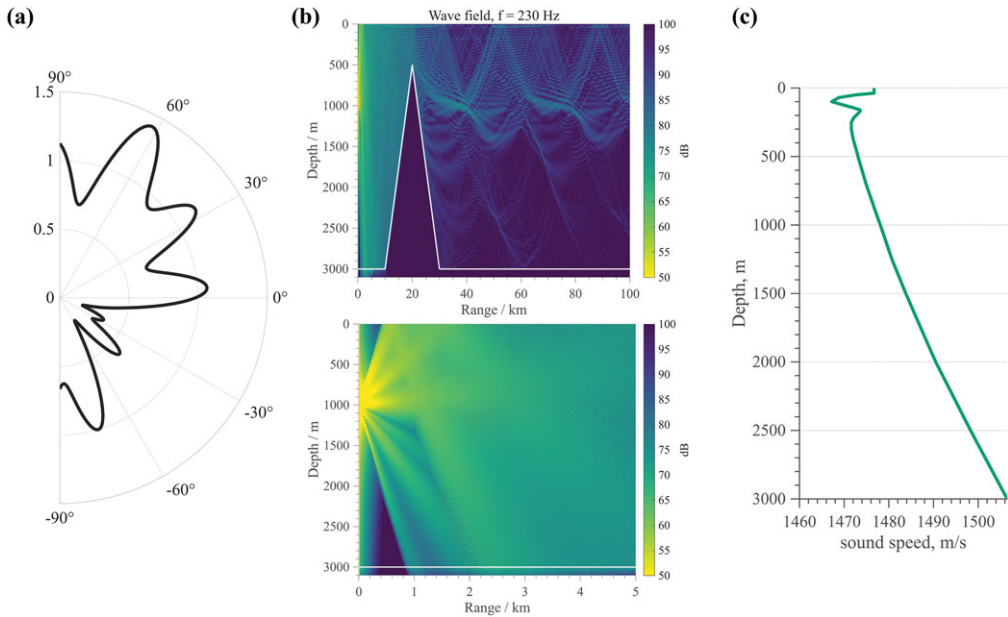


Figure 12. Directional underwater acoustic propagation across the Dickens seamount: (a) source directivity as a function of the grazing angle. (b) Sound field produced by the directional source. (c) North-Pacific sound-speed profile.

directivity shown in figure 12a is observed; however, its energy decays rapidly during upslope propagation. Consequently, the long-range propagation is dominated by modal interference. These results demonstrate the capability of multimodal PE to model underwater acoustic propagation from sources with prescribed directivity.

(d) Sound field in the presence of multi-layer seabed

Finally, we demonstrate the capability of multimodal PE to predict sound propagation in shallow water overlying a multi-layered seabed. In this case, a four-layer wedge model was simulated, in which the lower interface of the first layer (at $z = h_2$) matches that of the ASA benchmark wedge. The remaining two interfaces, located at $z = h_3$ and $z = h_4$, are defined by the linear functions $250\text{ m} - 0.0375 \times r$ and $300\text{ m} - 0.025 \times r$, respectively. The source was positioned where the water depth is 200 m, at a depth of 100 m. The remaining simulation parameters are summarized in table 1.

For comparison purposes, a FE simulation was conducted using only the pressure acoustics module implemented in the commercial software COMSOL. The detailed meshing scheme follows the recommendations in [11] and is omitted here for brevity. Figure 13 compares the results obtained by the two methods, showing strong agreement in both the water column and the multi-layer sediment. Minor discrepancies can be observed in the near-field region, where multimodal PE provides solutions with a limited propagation angle. Additionally, the FEM solution exhibits stronger interference patterns in the bottom layer, which may be attributed to the one-way approximation used in multimodal PE. A more detailed comparison is presented in figure 14, where the TL at a receiver depth of 30 m is plotted. The excellent agreement with the FEM prediction confirms the accuracy of multimodal PE for multilayer seabed scenarios, where multiple density discontinuities at stratified interfaces must be addressed simultaneously. This aspect is crucial for geoacoustic inversion, where an accurate forward model capable of dealing with complex sediments is required.

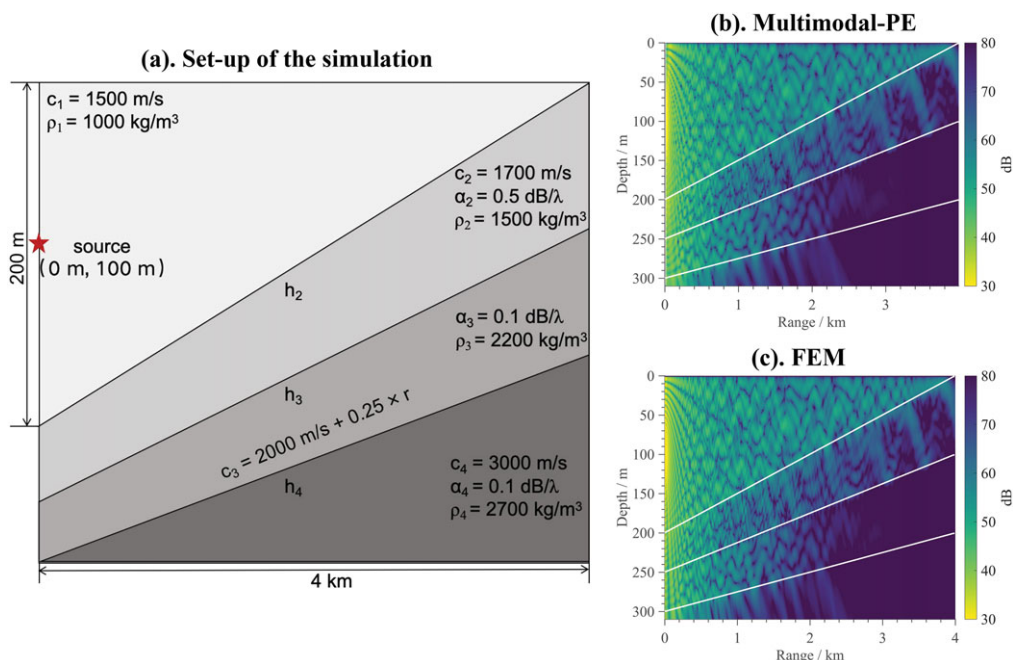


Figure 13. Sound propagation in the four-layer shallow water wedge, with the corresponding schematic shown in (a). (b–c) Illustrate the sound fields at 100 Hz predicted by multimodal PE and the benchmark FEM, respectively. Note that the FEM provides the solution covering full propagation angles.

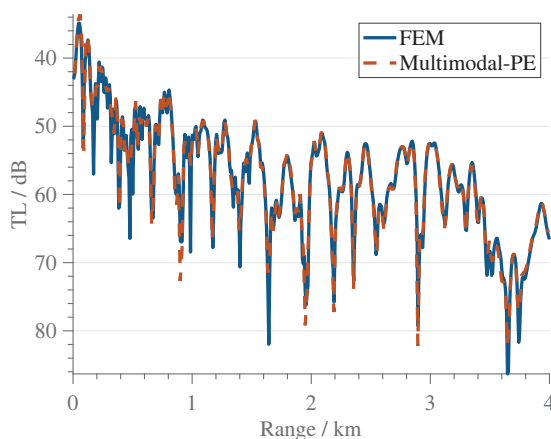


Figure 14. TL comparison at a receiver depth of 30 m at 100 Hz. The solid blue line represents the FEM result while the dashed red line denotes the multimodal PE solution.

4. Discussion

(a) Comparison with existing PE starters

So far, the multimodal PE method, together with its analytical starter, has been validated for modelling sound propagation from both point and directional sources, as well as in multi-layered seabed environments. It is essential to compare the proposed analytical starter with existing ones. Here, we consider two well-known analytical starters for comparison: Greene's starter [46] and

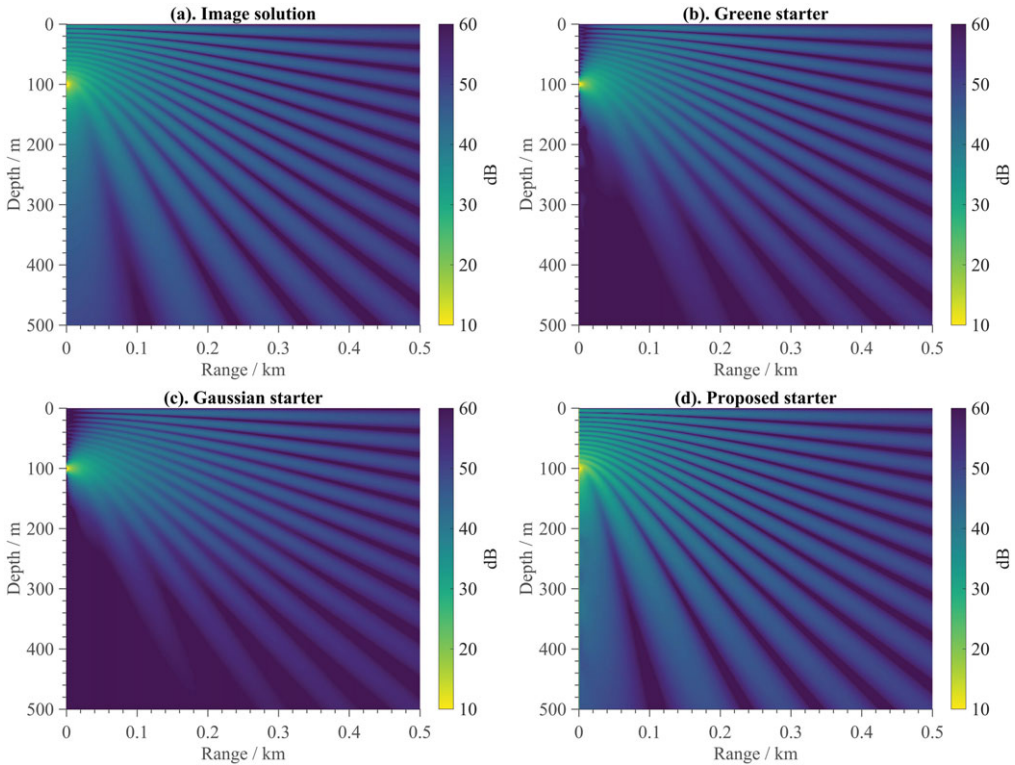


Figure 15. Sound propagation in the near-field region within 200 m, computed by (a) the reference FEM and multimodal PE with different starters. (b–d) Show results for the Greene, Gaussian, and proposed starters, respectively.

the Gaussian starter [47]. To be compatible with the multimodal PE framework, both starters must be projected onto the basis modes before practical implementation.

To clearly demonstrate the wide-angle capability of the proposed starter, we consider a point source located beneath a free surface, a classic scenario representing Lloyd mirror effects. The reference solution is derived analytically using the image method, assuming a source depth of z_s :

$$p = \frac{e^{ikR}}{R} - \frac{e^{ikR'}}{R'}, \quad (4.1)$$

where k is the wavenumber, and $R = \sqrt{r^2 + (z - z_s)^2}$, $R' = \sqrt{r^2 + (z + z_s)^2}$. To emphasize wide-angle propagation, the Padé approximation order is set to 16, and the number of basis modes is increased to $1.5M_c$ for all starters. The step size is set to 1 m.

Figure 15 compares the results obtained from the reference solution and the multimodal PE method with different starters. The source frequency is 100 Hz, and the source depth is $z_s = 100$ m. Compared with the image solution, the Gaussian starter exhibits the narrowest propagation angle, followed by Greene's starter. The proposed starter achieves nearly $\pm 90^\circ$ propagation. A more detailed comparison is presented in figure 16, which plots the TL as a function of depression angle relative to $z = 0$ for a fixed propagation range of 500 m. From these results, it is evident that the Gaussian and Greene starters produce effective propagation up to approximately 20° and 40° , respectively, without significant error. In contrast, the proposed starter supports extremely wide-angle propagation, nearly covering 90° . These findings demonstrate that the proposed analytical starter enables much wider-angle propagation than existing methods and is highly compatible with the multimodal PE framework. This capability is critical for modelling directional propagation, as beams travelling at large grazing angles require accurate wide-angle representations to capture wave behaviour appropriately. It is worth noting that the ray-based

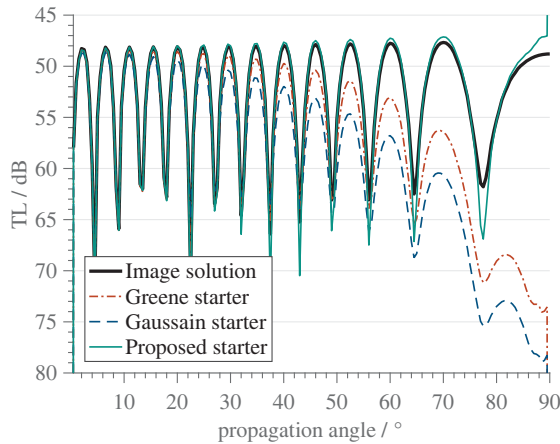


Figure 16. TL as a function of depression angle computed using various starters.

starter [40] is as simple to implement as the Greene and Gaussian starters, yet offers substantially improved wide-angle performance. This approach could also underpin a directional starter for underwater acoustic propagation, which will be investigated in future work.

(b) Relation to existing PE models

As one of the efficient approaches for modelling wide-angle propagation, the split-step Padé solution has been widely adopted for solving the PE in the underwater acoustics community. By integrating this method with normal modes, the so-called mode PE techniques have been developed to address three-dimensional problems [48–50]. In general, PE methods based on the split-step Padé scheme are implemented in the spatial domain, with the associated depth operator discretized in the z domain. The proposed multimodal PE can be categorized as a split-step Padé-based method, but its operator is constructed in the projection space (k_z domain). This formulation offers distinct advantages in handling strong contrasts in sound speed and density at stratified interfaces, as well as enabling very-wide-angle directional propagation, as demonstrated in this study.

On the other hand, the split-step Fourier (SSF) algorithm [10,51] provides an efficient alternative for implementing the PE. In this approach, the marching procedure is initiated by applying a Fourier transform to the sound pressure field, converting it from the spatial domain to the k_z domain. The field is then propagated in the k_z domain using a spectral propagator (similar to the angular spectrum method), which accounts for wave diffraction. Subsequently, an inverse Fourier transform returns the field to the spatial domain, followed by the inclusion of refraction effects through ζ^2 . The similarity between SSF and multimodal PE lies in the interpretation of the projection in multimodal PE as $\mathcal{F}\{\cdot\}$, which functions analogously to a sine transform from the spatial domain to the k_z domain. However, unlike SSF, multimodal PE does not split diffraction and refraction effects. Instead, it propagates $\mathcal{F}\{p(r, z)\}$ directly in the k_z domain, with both diffraction and refraction inherently incorporated via the eigenvalue k_r . This is evidenced by the eigenvalue decomposition of the depth operator $\mathbf{A}^{-1}\mathbf{B}$, which yields the eigenvalues of the normal modes Υ . Consequently, the multimodal PE formulation can be expressed as:

$$p(r, z) = \mathcal{F}^{-1} \{ e^{i\Delta r [\sqrt{k_r^2 + k_0^2} - k_0]} \mathcal{F}\{p(r_0, z)\} \}. \quad (4.2)$$

This formulation clearly demonstrates that diffraction and refraction are taken into account simultaneously during the marching algorithm. However, it should be noted that the SSF method is typically suited to narrow-angle propagation scenarios with negligible bottom interactions.

Moreover, strong contrasts in sound speed and density at the water–seabed interface can significantly degrade its computational efficiency. These limitations are effectively overcome by the proposed multimodal PE approach, which supports wide-angle propagation and analytically handles sound-speed and density discontinuities at stratified interfaces.

(c) Numerical complexity

Since most integrals in $\mathbf{C}$ and $\mathbf{A}$ are evaluated analytically, constructing these two matrices requires approximately $O(M^2)$ memory. The matrix $\mathbf{E}$, however, involves numerical integration and matrix multiplication, resulting in a computational cost of approximately $O(M^2) + O(N \log N)$. To construct $\mathbf{E}$ accurately, sufficient depth sampling is necessary. In this study, we compute the depth integrals using Clenshaw–Curtis quadrature, which yields a computational cost of $O(N \log N)$ according to the algorithm described in [52]. The inversion of $\mathbf{A}$, followed by matrix multiplication (i.e. $\mathbf{A}^{-1}\mathbf{B}$), incurs a cost of $O(M^3)$. Therefore, the overall computational cost of constructing the depth operator in multimodal PE is approximately $O(M^3)$.

The Clenshaw–Curtis quadrature rule typically requires only a few points per period of the basis modes to achieve high convergence accuracy. In this paper, we use five points per period, implying that $N = 2.5M$, given $M = 2Hf/c_0$. In general, M is substantially smaller than the number of depth discretization points typically used in finite-difference or finite-element methods. Additionally, at each range step, the Padé approximation requires $O(n_p M^2)$ operations, where n_p is the Padé order.

An important consideration is the updating of the depth operator. If the range dependence is limited to the sound speed, only $\mathbf{E}$ needs to be recomputed at each range step. However, if the bathymetry varies with range, all matrices must be updated as the propagator advances. The numerical integration required to construct the PML, corresponding to the bottom part of each matrix, is computationally less expensive than the main integral components of $\mathbf{C}$, $\mathbf{E}$, and $\mathbf{A}$, owing to its small thickness relative to the physical domain, typically only several acoustic wavelengths. Once initialized, these PML matrices remain unchanged as the propagator advances.

It is also worth noting that increasing the number of layers in the medium does not increase the computational complexity for a fixed H (deepest point of the computational domain). As seen in equations (2.37)–(2.38), no additional computational effort is required to handle density discontinuities at stratified interfaces. The integral over the entire depth is simply partitioned into sub-integrals, each corresponding to a layer. The total numerical cost remains equivalent to that of a single integral from $z = 0$ to $z = H$.

5. Conclusion

This article has presented multimodal PE, a multimodal formulation of the PE method. Unlike traditional spatial-domain implementations, multimodal PE projects sound pressure onto basis modes, deriving PEs for the corresponding projection coefficients together with the depth operator in the projection space. This approach offers three key advantages: (1) a stable solution that effectively handles discontinuities in sound speed and density at stratified interfaces and can be readily extended to multi-layer seabed scenarios; (2) strong convergence at a rate of $O(1/M^2)$ where M denotes the number of basis modes, which improves computational efficiency under high-accuracy demands; and (3) the capability to derive an analytical starter for sources with arbitrary directivity, including wide-angle propagation.

Numerical results validate the accuracy and efficiency of the proposed multimodal PE. Simulations demonstrate its ability to model sound propagation from directional sources, as confirmed by tests involving double pistons. The solutions were compared with those from RAM using a weighted modal starter. Results show that multimodal PE achieves stable solutions with coarse vertical discretization and fewer basis modes, whereas RAM requires careful design of the starter for reliable performance especially when beams with large grazing

angles and downsloping propagation are considered. The multi-layer seabed simulation, which involves multiple density discontinuities, further confirms that multimodal PE effectively handles interface discontinuities. Notably, increasing the number of layers does not increase the computational complexity when the overall depth of the domain remains fixed. In addition, the very-wide-angle propagation capability (almost covering $\pm 90^\circ$) of multimodal PE has been demonstrated through comparisons with analytical starters. Future work will aim to extend the multimodal formulation to three-dimensional PE models.

Data accessibility. The MATLAB code supporting this study, including scripts and example input files required to reproduce the key results and figures reported in the manuscript, is openly available at <https://github.com/TengjiaoHE/Multimodal-PE>. An archived, citable version is available via Zenodo at [53].

The data are provided in the electronic supplementary material [54].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. T.H.: conceptualization, formal analysis, funding acquisition, investigation, methodology, software, validation, visualization, writing—original draft; W.G.: investigation, methodology, supervision, writing—review and editing.

Both authors gave final approval for publication and agreed to be held accountable for the work performed therein.

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Appendix A. Eigenproblem for the normal mode $\Upsilon(z)$

The normal mode $\Upsilon(z)$ is determined by solving the Sturm–Liouville problem

$$\frac{d^2 \Upsilon_m}{dz^2} + k^2(z) \Upsilon_m = k_{r,m}^2 \Upsilon_m, \quad (\text{A } 1)$$

subject to the boundary conditions

$$\left. \begin{aligned} \Upsilon_m(z)|_{z=0} &= 0, \\ \text{and} \quad \frac{1}{\rho_1} \frac{d\Upsilon_m(z)}{dz} \Big|_{z=l^+} &= \frac{1}{\rho_2} \frac{d\Upsilon_m(z)}{dz} \Big|_{z=l^-}, \end{aligned} \right\} \quad (\text{A } 2)$$

The normal mode $\Upsilon(z)$ is then projected onto the basis modes $\Phi(z)$ using

$$\Upsilon^T = \Phi^T \mathbf{W}, \quad (\text{A } 3)$$

where $\mathbf{W}$ is the projection matrix that maps Υ onto Φ . To solve the Sturm–Liouville problem in equation (A 1), equation (A 1) is first multiplied by Φ , and the resulting weighted eigenequation is integrated over the waveguide depth. Applying the projection in equation (A 3) leads to

$$\int_0^H \frac{1}{\rho_j} \Phi \left[\frac{d^2}{dz^2} + k^2(z) \right] \Phi^T d\hat{z} \mathbf{W} = k_r^2 \int_0^H \frac{1}{\rho_j} \Phi \Phi^T d\hat{z} \mathbf{W}, \quad j=1,2. \quad (\text{A } 4)$$

In this equation, the integrals on the left- and right-hand sides correspond to the matrices $\mathbf{B}$ (defined in equation (2.22)) and $\mathbf{A}$ (defined in equation (2.20)), respectively. Consequently, the Sturm–Liouville problem in equation (A 1) is transformed into a generalized matrix eigenproblem through the projection in equation (A 3)

$$\mathbf{B}\mathbf{W} = k_r^2 \mathbf{A}\mathbf{W}. \quad (\text{A } 5)$$

It is now evident that the eigenvalues k_r can be obtained by performing the eigenvalue decomposition of the matrix $\sqrt{\mathbf{A}^{-1}\mathbf{B}}$, which in turn yields the projection matrix $\mathbf{W}$ assembled from the corresponding eigenvectors. Once $\mathbf{W}$ is determined, the normal modes Υ can be reconstructed through the projection defined by equation (A 3).

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